

PicoCTF 2026 – Timeline 0

Timeline 0

Forensics Medium 100 pts

by LT 'syreal' Jones

Can you find the flag in this disk image? Wrap what you find in the picoCTF flag format.

Download the disk image [here](#) ↗.

In this challenge, they want us to find the flag in the disk image

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Disk Image Forensics /w Sleuth Kit

```
└─$ tsk_gettimes partition4.img > body.txt
```

We can use The Sleuth Kit to check out the disk image

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The Sleuth Kit – tsk_gettimes Function

```
└─$ tsk_gettimes partition4.img > body.txt
```

```
└─$ cat body.txt
0|home|64770|d/drwxr-xr-x|0|0|1024|1762891277|1762891277|1762891277|1762891277
0|home/ctf-player|64771|d/drwxr-sr-x|1000|1000|1024|1762891182|1762891182|1762
0|lost+found|11|d/drwx————|0|0|12288|1762891277|1762891277|1762891277|176289
```

The Sleuth Kit `tsk_gettimes` function will output data that has MACtime data that can be used to track a timeline of file activity

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The Sleuth Kit – Mactime Function

```
└─$ mactime -b body.txt -d > timeline.csv
Old package separator "'" deprecated at /usr/bin/mactime line 154.
Old package separator "'" deprecated at /usr/bin/mactime line 167.

[the shy hat @ hackerfrags] - [/tmp/hf/time0]
└─$ cat timeline.csv | head
Date,Size,Type,Mode,UID,GID,Meta,File Name
Tue Jan 01 1985 07:00:00,41,macb,r/rrw-r--r--,0,0,4945,"bin/bcab"
Mon Oct 18 2021 07:54:17,451,ma .. ,r/rrw-r--r--,0,0,64994,"usr/share
```

The Sleuth Kit mactime function is used to create a timeline of all the files in the input, sorted by date and time

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```
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Old package separator "'" deprecated at /usr/bin/mactime line 167.

[theonlyhat@hackerfrags] - [/tmp/hf/time0]
└─$ cat timeline.csv | head
Date,Size,Type,Mode,UID,GID,Meta,File Name
Tue Jan 01 1985 07:00:00,41,macb,r/rrw-r--r--,0,0,4945,"bin/bcab"
Mon Oct 18 2021 07:54:17,451,ma .. ,r/rrw-r--r--,0,0,64994,"usr/share
```

This points us at a file with an odd timestamp, which was modified over two decades before all the other files on the filesystem

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```

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The Sleuth Kit – Icat Function

```
$ icat partition4.img 4945  
NzFtMzExbjNfMHU3MTEzc19oM3JfNDNhMmU3YWYK  
  
(theshyhat@hackerfrogs)-[/tmp/hf/time0]  
$ icat partition4.img 4945 | base64 -d  
71m311n1_0u7113e_b3e_41s2e7af
```

We can then use the Icat command to read that file and retrieve the flag